Project Name: ZeroFootPrint

Course Code: MTM4186

Team Number: Team 1

Member Name: Rahaf Kuveder

The Problem:

University campuses generate massive amounts of carbon emissions and waste due to the continuous, repetitive purchasing of new academic materials such as textbooks, engineering tools, lab coats, and electronics every semester. Manufacturing and shipping these items heavily accelerate global warming.

The Target User:

University students suffer from high financial expenses when buying new equipment, while other students have the exact same items sitting completely unused in their rooms.

Evidence and Statistics:

Studies show that producing a single new textbook emits around 7.5 kg of CO_2, and electronic devices generate even more. By swapping 3 items per student, a single campus can prevent tons of greenhouse gas emissions annually, directly contributing to a circular economy.

Why Traditional Apps Fail:

1. Lack of Trust and Safety: They lack secure campus only verification, making students hesitant to meet strangers.
2. High Costs and Commercial Focus: They focus on financial profit and charging fees rather than solidarity.
3. No Environmental Motivation: They completely lack awareness or interactive features that show students the ecological value of recycling.

ZeroFootPrint Solution:

Our app introduces Gamification with the earth timer to instantly reward students with visible environmental impact data, turning sustainability into an engaging campus movement.

Key Features:

Secure Campus Authentication: User login restricted strictly to university emails.

Campus Marketplace: A clean, optimized interface displaying available items categorized by type: Books Electronics, or Other.

Contact: Contacting the product provider via Email or WhatsApp.

Earth Countdown Counter: A live counter showing the estimated survival years left for Earth starting at 80 years, which actively increases whenever an item is swapped.

Offline Local Caching: Allows Students to browse viewed Campus items without an active internet connection.

The application strictly follows the MVVM architectural pattern to ensure clean, maintainable, and scalable code.

Model: Represents the structure of data.

View: Built using modern Flutter widgets Welcome Countdown Screen, Sign Up Screen, and Action Selection Screen.

View Model: Handles the backend logic and states without cluttering the UI code.

State Management Choice: We choose Provider or River pod as our primary state management tool to safely handle data flow and instantly notify views to re-render when a student clicks “Swap”.

Weekly Plan:

Week 2	Project Name decision
Week 3	Installing flutter and dart
Week 4	Learning Dart
Week 5	Learning Dart
Week 6	Learning Flutter
Week 7	Milestone 1
Week 8	Coding
Week 9	Coding
Week 10	Presentation
Week 11	Presentation
Week 12	Milestone 2

Figma Access Link:

https://www.figma.com/design/7KGfsYmCkvw6Ic2hCwpI2C/Untitled?node-id=0-1&t=IoqFlWjgTHwj1GT0-1

GitHub Repository Link:

https://github.com/rahafkuveder-hub/ZeroFootPrint-